

SUPPLEMENTARY MATERIAL

Computational study of structural and functional effects of EGFR nsSNPs and ncSNPs in glioblastoma

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Abstract

This supplement outlines the selection and analysis of coding and non-coding *EGFR* variants, including structural modeling, docking, molecular dynamics simulations, and clinical and regulatory analyses. It provides the decision rules, coordinate conventions, simulation settings, and statistical summaries used in the study. Supplementary Figures S1–S5 show sequence annotations, interaction networks, structural clusters, docking contacts, and molecular dynamics results. The accompanying supplementary workbook contains variant-level results and model assessments.

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1 Data sources and coordinate conventions

1.1 Reference sequence and coding variants

The human EGFR reference sequence was obtained from UniProtKB (P00533; 1210 amino acids). Coding variants were collected from dbSNP using GRCh37.p13 and the canonical protein identifier ENSP00000275493. The initial dataset contained 83,047 variants, including 5,909 missense variants. Comparison with the TCGA-GBM PanCancer Atlas mutation data identified 51 unique somatic missense substitutions for computational assessment.

Amino acid substitutions are reported according to their positions in the reference EGFR precursor sequence. The genomic position, reference and alternate alleles, protein substitution, and available variant identifiers were retained during annotation. Different substitutions at the same residue were treated as separate variants. For structural analyses, residue positions were matched to the numbered coordinates of the corresponding protein model.

1.2 Non-coding variants and genome assemblies

Non-coding somatic variants were obtained from the GRCh37 non-coding variant dataset in COSMIC version 103. Filtering for *EGFR* and central nervous system (CNS) tissues identified 45 allele-specific variants at 43 genomic positions, annotated as intronic, upstream, downstream, or splice-associated. Substitutions at the same position but with different alternate alleles were retained separately. One deletion was excluded from single-nucleotide screening, leaving 44 substitutions at 42 positions for AlphaGenome analysis.

GRCh37 served as the common coordinate system for tracking and reporting variants. Coordinates were converted to GRCh38 for tools that required this assembly, including AlphaGenome and Enformer. The resulting annotations and predictions were then linked to the original GRCh37 variant identifiers. GRCh37 was also used for motif sequence extraction. The corresponding positions of the two nearby candidate regulatory substitutions were as follows:

Substitution	GRCh37 position	GRCh38 position
T>G	chr7:55259509	chr7:55191816
T>A	chr7:55259524	chr7:55191831

These positions are reported as one-based genomic coordinates. The local Enformer analysis used zero-based, end-exclusive intervals for sequence extraction, with a one-base adjustment to locate each substitution within the extracted sequence. The two candidate loci were evaluated independently.

2 Coding variant prioritization

2.1 Predictor groups and consensus rules

The 51 missense variants were evaluated using three groups of predictors. Group 1 comprised seven tools that assessed deleteriousness, Group 2 comprised seven tools that assessed disease association, and Group 3 comprised five additional predictors used for cancer-related prioritization. A variant passed Groups 1 and 2 if at least four tools in each group classified it as damaging. It

passed Group 3 if at least three tools returned a positive prediction. Variants were retained only if they met the criteria for all three groups.

The classification criteria are listed below. Scores that could not be interpreted were recorded as unknown and did not count as positive votes. The required number of positive votes remained fixed, regardless of the number of predictions available for a variant.

Predictor	Criterion used for a positive vote
Group 1: deleteriousness; at least 4 of 7	
SIFT	Score < 0.05
PolyPhen-2 HVAR	Score > 0.908
PROVEAN	Score < -2.5
CADD	PHRED-scaled score > 20
REVEL	Score > 0.5
MutationTaster	Disease-causing or disease-causing automatic classification (D or A)
MutationAssessor	Score > 1.93
Group 2: disease-associated predictions; at least 4 of 7	
MetaSVM	D classification in the annotated prediction field
MetaLR	D classification in the annotated prediction field
AlphaMissense	Score > 0.56
SuSPect	Score $\geq$ 50
PHD-SNPg	Pathogenic classification
MetaSNP	Disease classification
VARITY	VARITY_R score > 0.5
Group 3: additional prioritization; at least 3 of 5	
FATHMM-XF	Coding score > 0.5
MetaRNN	Score > 0.5
VEST4	Score > 0.5
BayesDel (addAF)	Score > 0.16
CScape	Score $\geq$ 0.5

Forty-two variants met the criteria for all three groups and were retained for further analysis. Agreement among predictors was used for prioritization and was not considered experimental evidence of pathogenicity or drug resistance. Supplementary Tables S1 and S2 describe the predictors and list the individual scores and classifications.

2.2 Protein stability assessment

PremPS, MAESTRO, and mCSM were used to estimate changes in protein stability associated with each substitution. The sign conventions of the tools were considered separately: positive $\Delta\Delta G$ values from PremPS and MAESTRO indicated destabilization, whereas negative values from mCSM indicated destabilization. A variant was retained if at least two of the three tools predicted a destabilizing effect.

This criterion retained 39 of the 42 variants. S123F, R252P, and A289D did not meet the stability criterion. Supplementary Table S3 provides the values from each tool and the resulting consensus classifications. Predictions from the three tools were treated as separate votes and were not combined into a mean free-energy estimate.

2.3 Spatial clustering and selection of structural targets

For each retained variant that could be mapped to the structure, the C_{α} coordinate was extracted from chain A of the wild-type EGFR model. Pairwise Euclidean distances were calculated, and average-linkage clustering was performed in SciPy. The dendrogram was cut at 25 Å, after also examining cutoffs of 10, 15, and 20 Å. The cluster summary included groups with at least two variant entries; singleton assignments were omitted.

The two largest reported clusters contained 16 and 12 substitutions, mainly in extracellular regions. The cluster numbers used in the presentation follow the order in Supplementary Table S4 rather than the labels generated by hierarchical clustering. A spatial cluster may include residues from more than one sequence domain. Supplementary Figure S3 shows the three groups selected for structural visualization and their positions in the sequence.

Within the two largest extracellular clusters, ten substitutions with PremPS $\Delta\Delta G > 1.25$ were selected: G63R, R108K, R222C, C240Y, D256G, A289V, P596R, G598V, C624F, and C636Y. The threshold was based on R108K, which had a PremPS value of 1.26. C240Y and G598V were selected for extracellular-domain modeling and antibody docking. V774M and L861Q were assessed separately as tyrosine kinase-domain targets in erlotinib docking and molecular dynamics simulations. The threshold used for the extracellular shortlist was therefore not applied as a general criterion for the kinase-domain targets.

3 Sequence annotation and structural models

3.1 Secondary structure, accessibility, and functional annotations

Secondary structure was predicted from the EGFR sequence using SOPMA and PSIPRED. NetSurfP-3.0 was used to obtain residue-level structural and solvent-accessibility predictions. Residues with relative solvent accessibility above 25% were classified as exposed, whereas those with values of 25% or less were classified as buried. Supplementary Figure S1 shows the wild-type profiles, and Supplementary Table S5 provides comparisons for the selected residues.

MusiteDeep was used to predict phosphorylation, ubiquitination, and N-linked glycosylation sites, while NetPhos 3.1 was used to predict serine, threonine, and tyrosine phosphorylation. GPS 6.0 was run with default settings for the AGC, CAMK, and tyrosine kinase families. The predictions were mapped to the EGFR sequence to identify variants located at or near potential modification sites. Proximity to a predicted site was recorded as an annotation and was not considered evidence of an altered modification.

Project HOPE was used to examine possible changes in residue size, charge, hydrophobicity, and local interactions. MutPred2 predictions were evaluated using an overall score threshold of 0.5, and associated molecular mechanisms were considered when the reported empirical value was $p \leq 0.05$. The selected phenotypic and mechanistic predictions are listed in Supplementary Table S6.

3.2 Interaction networks

EGFR was used as the query for functional association analysis in GeneMANIA and protein interaction analysis in STRING. GeneMANIA integrated evidence from physical interactions, co-expression, co-localization, genetic interactions, pathways, and shared domains. STRING interactions were evaluated by confidence score, with scores of at least 0.90 classified as high confidence. K-means clustering was used to group the interacting proteins. The exported network views are shown in Supplementary Figure S2.

3.3 Model generation and quality assessment

Wild-type and mutant domain structures were generated with SWISS-MODEL using the AlphaFold EGFR structure AF-P00533-F1-v6 as the template. The 620-amino-acid extracellular-domain models represented wild type, C240Y, and G598V. The 268-amino-acid kinase-domain models represented wild type, V774M, and L861Q. Each mutant model contained one amino acid substitution.

Model quality was evaluated using the SWISS-MODEL assessment suite, QMEANDisCo, MolProbity, ERRAT, PROCHECK through SAVES v6.1, and ProSA. Reported measures included Ramachandran distributions, PROCHECK core-region occupancy, and ProSA Z-scores. Supplementary Table S7 lists these results. TM-align and ChimeraX were used to compare fold similarity and coordinate deviations between each mutant model and its wild-type domain. These comparisons apply to the generated models and do not represent conformational sampling in solution.

4 Molecular docking

4.1 Preparation of receptors and binding partners

Protein structures were prepared in ChimeraX by adding hydrogen atoms and assigning charges. The extracellular-domain models were paired with cetuximab or nimotuzumab to form six antibody–receptor systems. The wild-type, V774M, and L861Q kinase-domain models were each paired with erlotinib. For small-molecule docking, Open Babel and the Universal Force Field were used for 500 steps of minimization. Receptor and ligand files were provided to AutoDock Vina in PDBQT format.

4.2 HADDOCK3 restraints and sampling

Ambiguous interaction restraints were used to guide antibody docking. For cetuximab, the active EGFR residues were 373, 376, 377, 379, 386, 408, 410, 411, 412, 414, 432, 433, 436, 441, 442, 464, 465, 467, 489, 491, 492, and 497. Active residues in the cetuximab light chain were 27, 32, 50, and 91–96, while those in the heavy chain were 31, 52–59, and 100–105.

For nimotuzumab, the active EGFR residues were 353, 356, 357, 358, and 359. Active residues in the light chain were 28, 34, 58, and 94–103, and those in the heavy chain were 28, 47, 52, 58, 97, and 101–107. The lists in Supplementary Table S8 use the residue numbering of the source chains. The prepared antibody chains and corresponding restraint files define the numbering used in each docking system. Surface residues within 6.5 Å of an active residue were assigned as passive residues using the HADDOCK3 `active_to_passive` utility.

The HADDOCK3 workflow generated 500 rigid-body models. Of these, 200 were selected

for semi-flexible refinement, and 100 refined models were retained for clustering based on the fraction of common contacts. The workflow also included caprieval evaluation stages, selection of representatives from the leading clusters, and contact-map analysis. These model counts refer to sampling within a single workflow and do not represent independent replicate experiments.

Selected models were evaluated using HADDOCK scores and their energy components, buried surface area, and interface contacts. PRODIGY was used to predict binding free energies and dissociation constants for the antibody complexes. Supplementary Table S8 contains summaries of the selected models and the active-residue definitions. HADDOCK scores and PRODIGY energies were reported separately.

4.3 AutoDock Vina search space and pose analysis

The kinase binding pocket was defined with reference to residues 718, 726, 743, 745, 790, 793, and 844. The AutoDock Vina search box measured $20 \times 20 \times 20$ Å and was centered at $x = -11.414$, $y = 33.957$, and $z = 12.442$ Å. Exhaustiveness was set to 32. The same search space was used for the wild-type and mutant kinase models.

Selected complexes were examined in ChimeraX and BIOVIA Discovery Studio Visualizer to identify hydrogen bonds, hydrophobic contacts, and differences in ligand orientation. The resulting two-dimensional interaction maps are shown in Supplementary Figure S4. The small-molecule section of Supplementary Table S8 reports the Vina docking scores and PRODIGY-Lig estimates separately. Because these values were obtained using different scoring procedures, they were not combined on a common energy scale.

5 Molecular dynamics simulations

5.1 System composition and parameterization

Molecular dynamics simulations were performed for the wild-type, V774M, and L861Q EGFR kinase domains in complex with erlotinib using GROMACS 2022.2. The protein was parameterized with AMBER99SB-ILDN. Erlotinib parameters were generated with ACPYPE/ANTECHAMBER using GAFF2 atom types, AM1-BCC charges, and a net ligand charge of zero. Each system was solvated in a cubic TIP3P water box with a minimum distance of 1.0 nm between the solute and the box boundary.

Each system contained one protein molecule and one erlotinib molecule. The wild-type, V774M, and L861Q systems contained 14,727, 14,728, and 14,719 water molecules, respectively, together with 45 sodium ions and 48 chloride ions.

5.2 Minimization and equilibration

Energy minimization was performed by steepest descent for a maximum of 50,000 steps, using a step size of 0.01 nm and a maximum-force target of $1000 \text{ kJ mol}^{-1} \text{ nm}^{-1}$. Position-restrained NVT and NPT equilibration each consisted of 50,000 steps at 2 fs, corresponding to 100 ps per stage. Initial velocities were generated at 310 K for the NVT stage. NPT equilibration continued from the NVT state without generating new velocities.

Temperature was controlled with the V-rescale thermostat using separate protein–ligand and water–ion coupling groups. Both groups were maintained at 310 K with coupling constants of

0.1 ps. NPT equilibration and production used isotropic Parrinello–Rahman pressure coupling at 1 bar, with a coupling constant of 2 ps and a compressibility of $4.5 \times 10^{-5} \text{ bar}^{-1}$.

5.3 Production settings and trajectory duration

Production simulations used the leap-frog integrator with a 2 fs time step and periodic boundary conditions in all three dimensions. The Verlet cutoff scheme was applied with short-range electrostatic and van der Waals cutoffs of 1.0 nm. Long-range electrostatic interactions were calculated with particle mesh Ewald using fourth-order interpolation and a Fourier spacing of 0.16 nm. Bonds involving hydrogen were constrained with LINCS using one iteration and order four. Long-range dispersion corrections were applied to both energy and pressure.

Compressed coordinates were saved every 50,000 steps (100 ps), while energies and log entries were saved every 5000 steps (10 ps). Production continued from the equilibrated systems without regenerating velocities. The completed trajectories were 500 ns for wild type and 200 ns for each mutant, giving a total production time of 900 ns. With a 2 fs time step, these durations correspond to 250 million and 100 million integration steps, respectively. Comparative analyses used the first 200 ns of the wild-type trajectory and the full 200 ns trajectory for each mutant.

EGFR–erlotinib system	Total production (ns)	Comparative window (ns)
Wild type	500	0–200
V774M	200	0–200
L861Q	200	0–200
Total	900	600

The extended wild-type trajectory provided additional sampling for that system. It was not treated as an independent replicate or as evidence that the mutant systems had converged.

5.4 Trajectory summaries and statistical interpretation

GROMACS analysis tools were used to calculate protein root mean square deviation (RMSD), residue root mean square fluctuation (RMSF), radius of gyration, solvent-accessible surface area (SASA), and protein–ligand hydrogen bonds. RMSD, RMSF, and radius of gyration were reported in nm, and SASA in nm^2 . The exported trajectory summaries were plotted in R using `ggplot2` and related packages.

Supplementary Figure S5 reports means and standard deviations. RMSD, radius of gyration, and SASA were summarized across sampled time points, whereas RMSF was summarized across the residue-level values in each profile. The pairwise annotations were generated using two-sided, unpaired `t.test` comparisons with the default unequal-variance formulation. No adjustment for multiple comparisons was specified. Because time points within a trajectory and residues within a protein are not independent simulation replicates, these annotations describe differences between sampled distributions rather than uncertainty across independent simulations.

6 Clinical expression, immune, and survival analyses

6.1 Expression data and patient matching

EGFR expression data were obtained from the UCSC Xena Pan-Cancer Atlas EB++AdjustedPANCAN RNA-sequencing matrix. Clinical annotations and overall survival data were obtained from the TCGA Clinical Data Resource. Positions 14–15 of the TCGA barcode were used to identify primary solid tumors (01), while normal solid tissue samples (11) were retained for baseline comparisons where applicable. Patients were matched using the first 12 barcode characters. If an identifier appeared more than once, the first record in file order was retained for patient-level analysis. Expression and clinical data were then joined by patient identifier.

Expression was analyzed on the scale provided in the normalized source matrix, without conversion to transcripts per million (TPM). Cancer-level summaries were calculated as median expression values and standardized across cohorts for the pan-cancer heatmap. The corresponding molecular summary used the cBioPortal alteration-frequency export. Reported frequency entries were summed within each study and standardized for visualization. This measure was treated as a descriptive molecular score, not as a patient-level joint probability of alteration.

6.2 Clinical categories and immune associations

Expression distributions were compared across pathological stages for carcinomas and histological grades for gliomas. In the combined glioma grade plot, GBM samples were assigned to G4, and LGG grade annotations were grouped into G2 and G3. Missing, unreported, or unrecognized categories were excluded from the relevant comparisons.

Primary-tumor expression values were matched to the Thorsson immune subtype classifications (C1–C6). Within each cancer type, samples with expression above the cohort median were classified as overexpressed for the immune-subtype summaries. Because a strict greater-than rule was used, samples equal to the median were not included in the overexpressed group. Associations with ESTIMATE, Immune, and Stromal scores were assessed by Pearson correlation using the matched samples available for each comparison.

6.3 Survival models

Overall survival status and time were obtained from TCGA-CDR. Records with missing survival or expression data, or with non-positive survival times, were excluded. Survival time in days was converted to months using a divisor of 30.4167. Estimates for the GBM and LGG forest plot were obtained from separate univariable Cox models using continuous EGFR expression as the predictor.

An additional LGG model included expression, age at diagnosis, and grade (G2 or G3) and was fitted using complete cases for these variables. Hazard ratios and 95% confidence intervals were calculated from the fitted Cox model. This adjusted LGG analysis was separate from the univariable comparison of GBM and LGG.

For the LGG Kaplan–Meier plot, values at or above the cohort median defined the high-expression group, and values below the median defined the low-expression group. The curves show survival in these two groups. Adjusted hazard ratios and significance estimates were obtained from the Cox model containing continuous expression, age, and grade.

7 Non-coding regulatory analyses

7.1 AlphaGenome screening

AlphaGenome was queried for each unique single-nucleotide variant using its converted GRCh38 position and a surrounding 16,384 bp interval. The requests included ATAC-seq and RNA-seq outputs without restriction by ontology term. The screening statistic was defined as the maximum absolute difference between alternate and reference ATAC predictions across all returned positions and tracks:

$$D_{\text{ATAC}} = \max_{i,t} |A_{i,t}^{\text{alt}} - A_{i,t}^{\text{ref}}|,$$

where i denotes position and t denotes track. This statistic does not identify a specific tissue or distinguish increased from decreased accessibility. It was used for prioritization and was not interpreted as a calibrated probability.

The screening threshold was $D_{\text{ATAC}} \geq 0.5$, which was met by six substitutions. The table lists their scores and indicates the five substitutions included in the subsequent motif analysis. Positions are reported in GRCh37.

GRCh37 variant	ATAC difference	Motif analysis
chr7:55238076 A>T	0.546875	Yes
chr7:55238084 C>T	0.55078125	Yes
chr7:55240707 G>A	0.552734375	Yes
chr7:55259509 T>G	0.500000	Yes
chr7:55259524 T>A	0.750000	Yes
chr7:55260454 C>A	0.500000	No

7.2 Regulatory annotation and motif comparisons

The five selected loci were annotated with RegulomeDB, HaploReg, and Ensembl VEP using the GRCh37 Regulatory Build. For each locus, a 31 bp forward-strand sequence containing 15 bases on either side of the variant was extracted from GRCh37. A corresponding alternate sequence was created by replacing the central base. The reference base obtained during sequence retrieval was checked against the supplied reference allele.

Reference and alternate sequences were scanned against JASPAR vertebrate position frequency matrices using an 85% relative-profile-score threshold. Exported motif results were grouped by variant, transcription factor, and sequence state. If a sequence contained more than one hit for a factor, only the highest relative score was retained. Scores reported on a 0–1 scale were converted to percentages before comparison.

For each factor, the score difference was calculated as $\Delta S = S_{\text{alt}} - S_{\text{ref}}$. If no hit was exported for one sequence state, its score was set to zero for the comparison. Changes of $|\Delta S| \geq 5$ percentage points were selected, along with motif appearances or disappearances for which the observed score exceeded 85%. A large difference can therefore result when a hit passes the reporting threshold in one sequence but is not exported for the other. It does not directly measure binding affinity. Motif matches were used to identify possible regulatory changes, not to establish transcription factor occupancy.

7.3 Chromatin context, conservation, and epigenetic annotation

The candidate loci were examined in the 3D Genome Browser 2.0 using the Glioblastoma_G523 dataset, which was derived from *in situ* Hi-C data from primary glioblastoma stem-cell cultures. The browser view covered the 350 kb GRCh38 interval chr7:54932307–55282307. TAD intervals, chromatin-loop calls in BEDPE format, and compartment PC1 values were reviewed to describe the EGFR genomic neighborhood.

Chromatin annotations were interpreted in their respective genome assemblies, while the original GRCh37 identifiers were retained for cross-reference with the variant table. The contact map, domain boundaries, and loop calls provided regional information about possible regulatory interactions. These data describe the chromatin context of the loci but do not measure substitution-associated changes in promoter contact.

Evolutionary conservation was assessed using base-resolution phyloP and phastCons annotations. The GRCh38 positions were also compared with ENCODE candidate cis-regulatory elements. The interval chr7:55190816–55192831 was examined to assess the local epigenetic context of both candidate loci. Available annotations included DNase/ATAC accessibility, H3K27ac, H3K4me1, H3K4me3, and CTCF. Regional signals were distinguished from peaks that directly overlapped the altered base.

7.4 Patient recurrence and expression-data queries

COSMIC non-coding records were linked to the Classification registry to map phenotype identifiers to patient identifiers. Recurrence was calculated after deduplication at the patient level. The pan-cancer denominator was the number of patients with an EGFR non-coding alteration in the queried records, not the total number of sequenced cancer patients. The CNS denominator was defined in the same way after filtering by tissue.

The pan-cancer dataset contained 8404 patients with an EGFR non-coding alteration, including 273 with chr7:55259524 T>A. The CNS subset contained 99 patients: three with chr7:55259524 T>A and one additional patient with chr7:55259509 T>G. The combined proportion was therefore 4/99 (4.04%) in this selected CNS non-coding cohort. This value does not estimate prevalence among all GBM or CNS tumors.

CADD PHRED-scaled scores and annotations were obtained for the candidate non-coding substitutions. TCGA, PCAWG, and CCLE/DepMap were searched for samples containing both the targeted variants and expression measurements. These searches did not identify matched mutant expression data suitable for comparison. Consequently, no measured mutant-versus-wild-type expression effect was estimated.

7.5 Enformer sequence construction and scoring

Enformer was run locally using a PyTorch implementation and pretrained model weights. The model configuration had a dimension of 1536, a depth of 11, eight attention heads, 896 output bins, and output heads for 5313 human and 1643 mouse tracks. Inference was performed on the CPU using the human output head, evaluation mode, and disabled gradient calculation.

A 196,608 bp sequence was extracted from GRCh38 chromosome 7 using the zero-based interval [55040964, 55237572), centered at 55139268. The window was selected so that both the EGFR transcription start site and the candidate regulatory region fell within the central prediction interval. Separate alternate sequences were created with the T>G substitution at one-based position 55191816 or the T>A substitution at position 55191831. Each sequence was

one-hot encoded before inference.

For each human track, the promoter summary was the maximum prediction over zero-based bins 36, 37, and 38, corresponding to the slice 36:39. Fold change was calculated as

$$F_t = \frac{\max_{b \in \{36,37,38\}} P_{b,t}^{\text{alt}}}{\max_{b \in \{36,37,38\}} P_{b,t}^{\text{ref}} + 10^{-4}}.$$

The pseudocount was added only to the reference denominator. Tracks with an alternate promoter signal above 0.5 were retained, and the track with the largest fold change was selected separately for each substitution. If no track met the signal threshold, the largest fold change across all tracks was used.

The reported values therefore correspond to a track selected by fold change after signal filtering, rather than to the maximum expression signal across all tracks. The human output tracks represent different assays and cellular contexts, and the predicted values are not TPM measurements. This local analysis does not directly test enhancer activity in glioblastoma.

8 Computational requirements and reproducibility

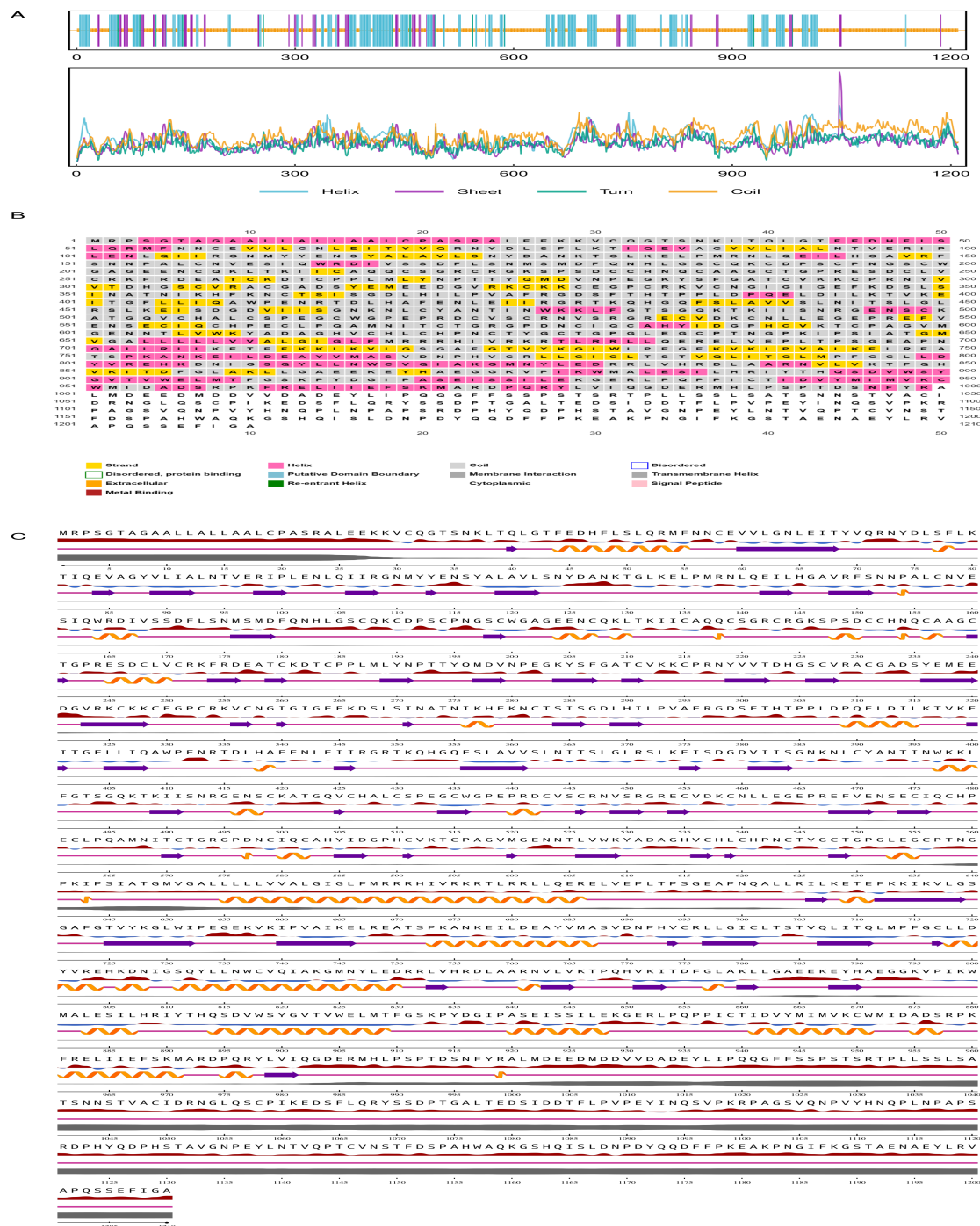
The analyses were performed on a laptop with a four-core central processing unit, 8 GB of random-access memory, and an NVIDIA GTX 1650 Ti graphics processing unit with 4 GB of video memory. Molecular dynamics production totaled 900 ns across the three EGFR–erlotinib systems. GROMACS used four OpenMP threads with GPU acceleration, whereas local Enformer inference was performed on the CPU. AlphaGenome and other web-based analyses required access to external services.

9 Supplementary table guide

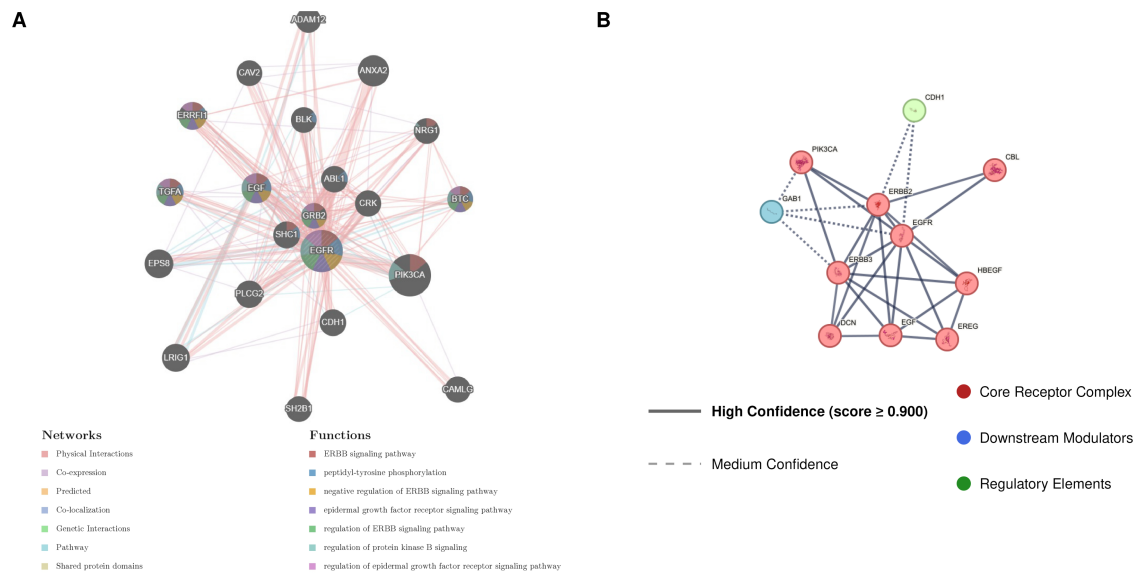
The accompanying `Supplementary.xlsx` workbook contains eight worksheets, named Table S1 through Table S8. Their contents are summarized below.

Worksheet	Contents
Table S1	Variant-effect predictors, score interpretations, and group-level consensus criteria.
Table S2	Individual predictor outputs for the EGFR missense variants.
Table S3	Stability predictions for the 42 selected variants and the stability consensus.
Table S4	Spatial clustering parameters, reported clusters, and the extracellular shortlist.
Table S5	Secondary structure and solvent accessibility at the selected variant residues.
Table S6	Predicted phenotypic and molecular effects of selected substitutions.
Table S7	Quality assessment and structural comparisons of the domain models.
Table S8	Selected antibody and erlotinib docking results, antibody active-residue definitions, and Vina search-space parameters.

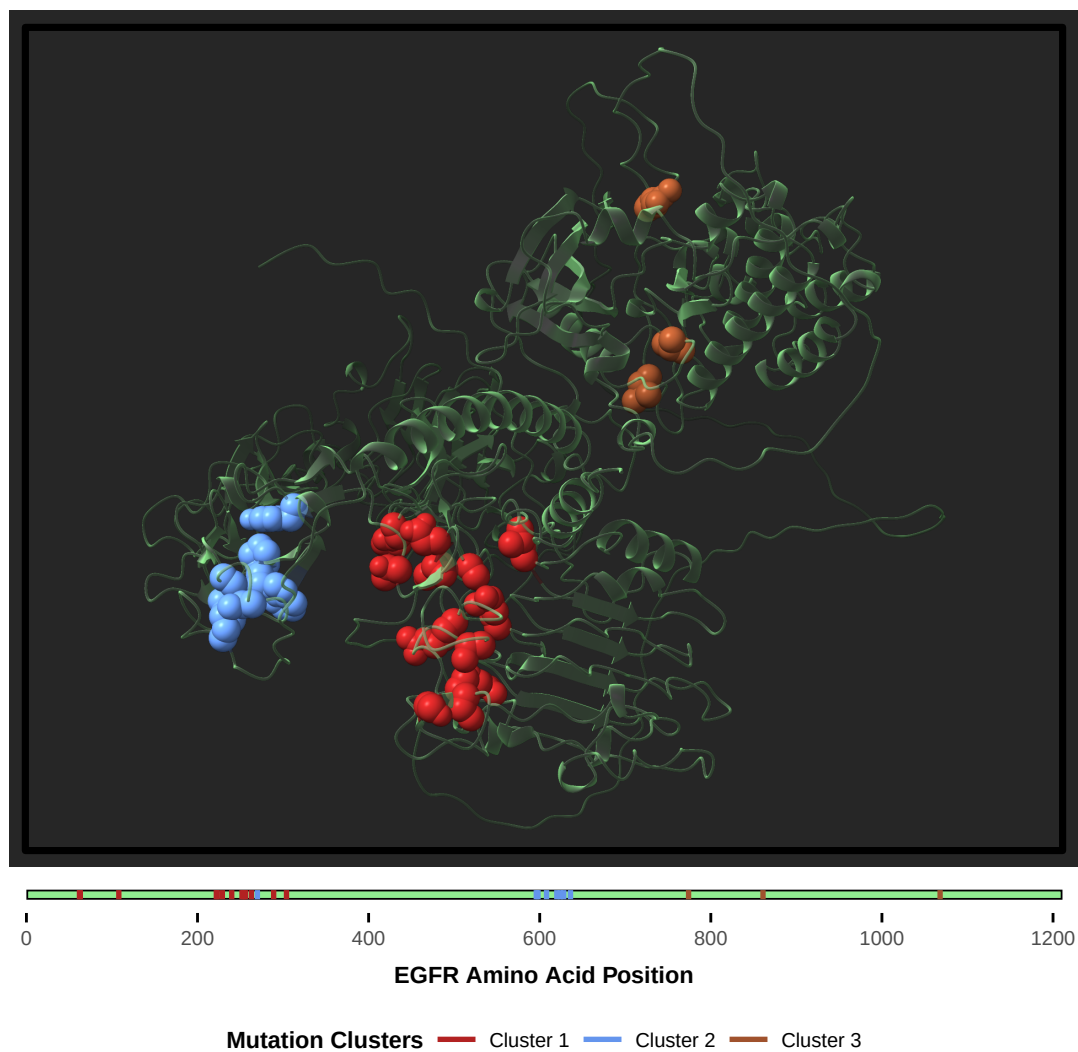
10 Supplementary figures



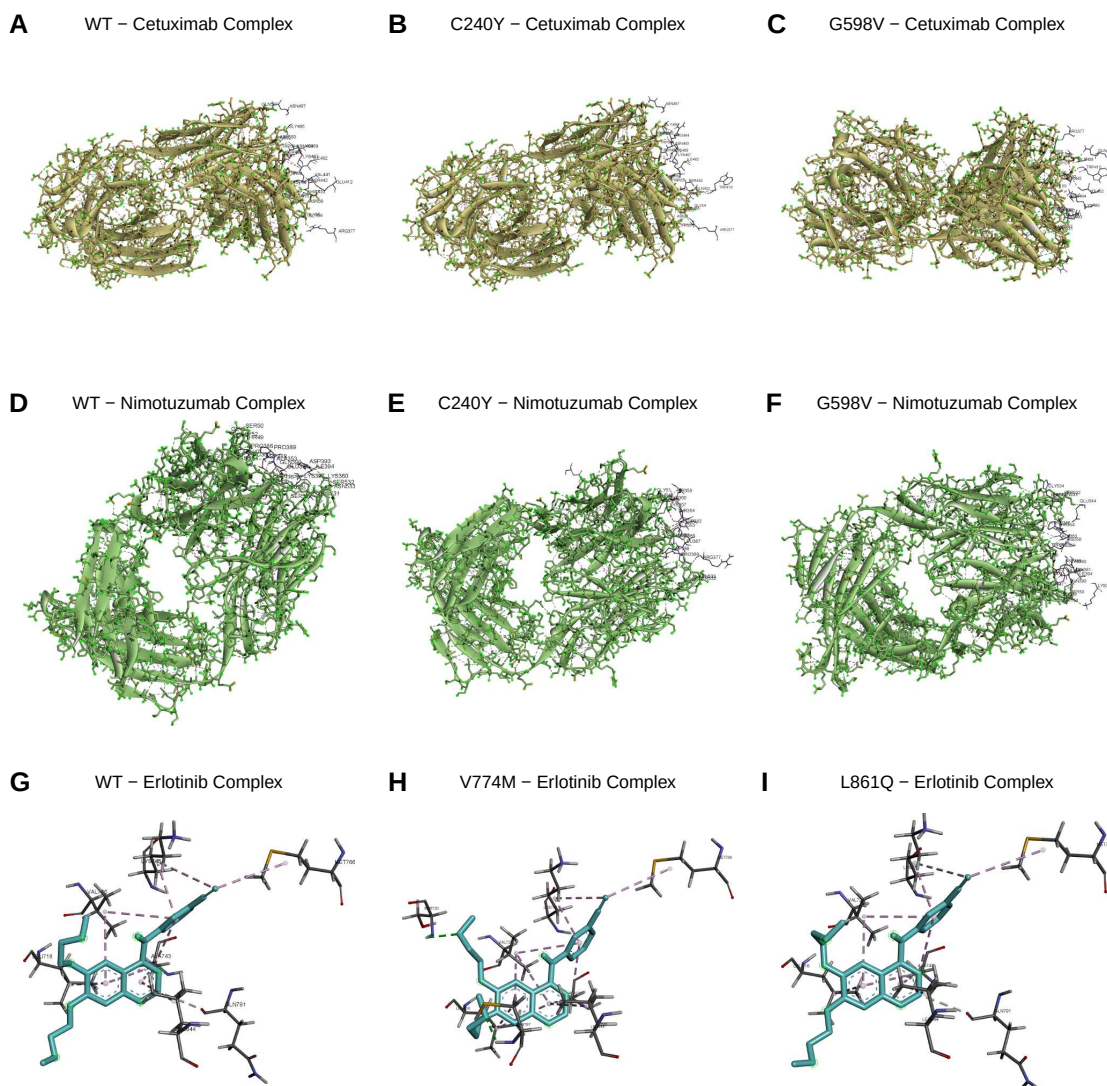
Supplementary Figure S1. Sequence-based structural annotation of wild-type EGFR. (A) SOPMA secondary-structure assignments and score profiles across the sequence. (B) PSIPRED sequence annotation with the corresponding structural and functional feature key. (C) NetSurfP-3.0 residue-level structural profiles. All panels show the full-length EGFR sequence. Comparisons for the selected variant residues are provided in Supplementary Table S5.



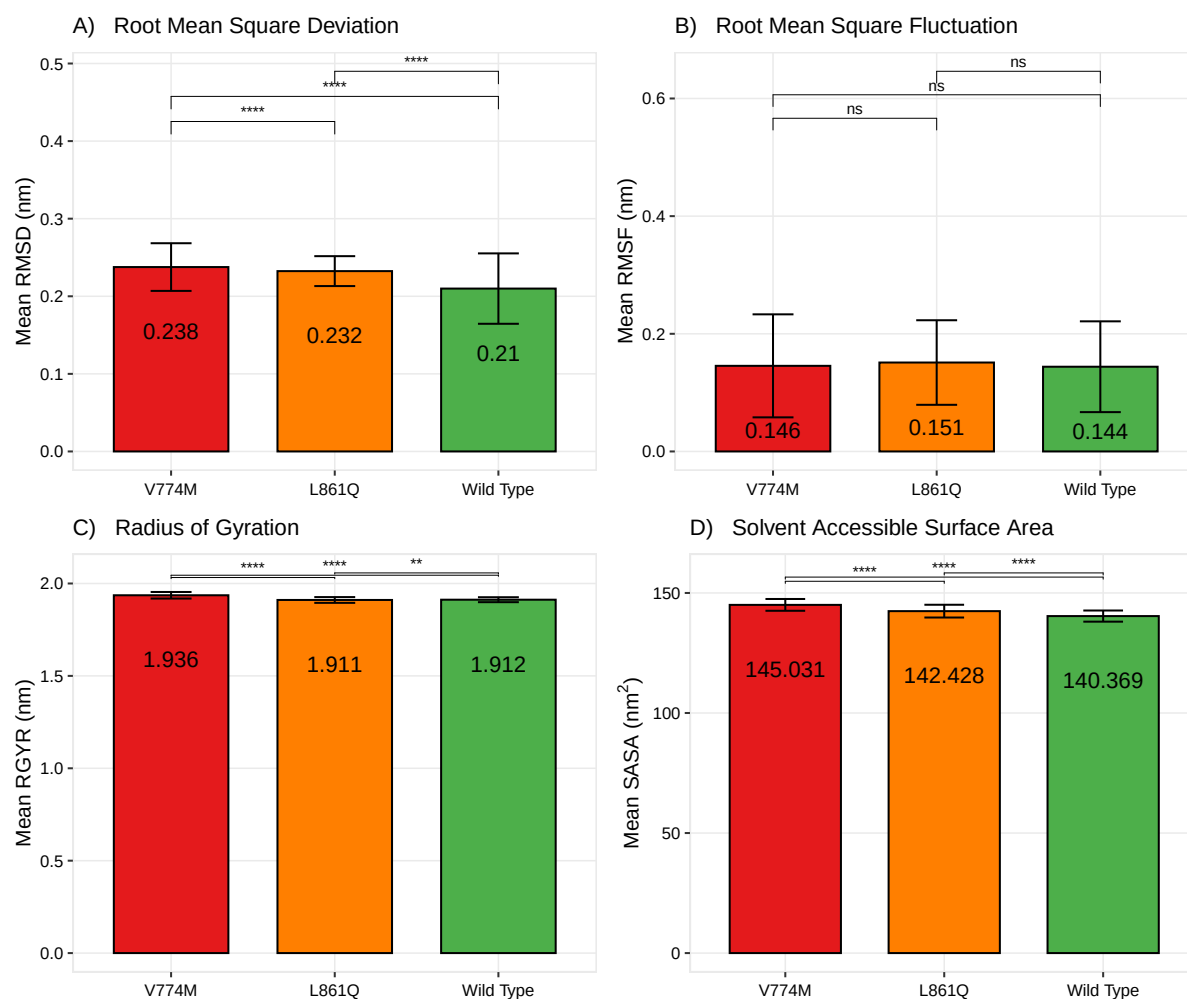
Supplementary Figure S2. Functional association and protein interaction networks for EGFR. (A) GeneMANIA network, with edge colors denoting evidence categories and node annotations denoting associated functions. (B) STRING interaction network with confidence and node-group legends. The networks show associations with EGFR and do not represent experimentally measured effects of individual variants.

**Supplementary Figure S3. Structural and sequence distribution of three selected mutation groups.**

The upper panel shows the EGFR structural model with the selected residue positions highlighted, and the lower panel maps these positions onto the 1210-residue sequence. Red, blue, and brown denote the groups labeled Clusters 1, 2, and 3, respectively. Different substitutions at the same residue share one sequence position. Supplementary Table S4 provides the full cluster summary.



Supplementary Figure S4. Two-dimensional interaction maps for the selected docked complexes. (A–C) Cetuximab complexes with the wild-type, C240Y, and G598V extracellular domains, respectively. (D–F) Nimotuzumab complexes with the same three extracellular-domain models. (G–I) Erlotinib complexes with the wild-type, V774M, and L861Q kinase domains, respectively. Contact types are defined by the keys within each panel. The maps represent selected docking poses and do not show contact persistence during molecular dynamics simulations.



Supplementary Figure S5. Molecular dynamics summaries for the matched 200 ns trajectories. (A) Protein RMSD, (B) residue RMSF, (C) radius of gyration, and (D) SASA for the wild-type, V774M, and L861Q EGFR–erlotinib systems. Bars and error bars represent means and standard deviations. Panels A, C, and D summarize sampled time points, whereas panel B summarizes residue-level fluctuations. Brackets show the existing unpaired, two-sided, unequal-variance *t*-test annotations (ns, $p > 0.05$; **, $p \leq 0.01$; ****, $p \leq 0.0001$). These comparisons do not account for temporal or spatial dependence and do not represent tests across independent simulation replicates.